

ESP32-S3 Smart Shoe Arduino Test Setup Guide

This document can be forwarded to a third party for testing the Arduino firmware and hardware setup for the ESP32-S3 smart shoe project.

1. Required Arduino packages and libraries

- 1 esp32 by Espressif Systems
- 2 NimBLE-Arduino
- 3 Built-in only, no separate install needed: Arduino.h, Wire.h, math.h
- 4 No separate MPU6050 / MS5611 / HMC5883L / QMC5883L libraries needed because the firmware uses direct register-level I2C access

2. Recommended Arduino IDE environment

- Arduino IDE 2.x
- ESP32 board package: esp32 by Espressif Systems 3.x
- NimBLE-Arduino 2.x

3. Recommended board settings

Setting	Recommended value
Board	ESP32S3 Dev Module
USB CDC On Boot	Enabled
CPU Frequency	240 MHz
Flash Size	Match the board, commonly 4 MB or 8 MB
Partition Scheme	Default / board default
PSRAM	Disabled unless the board definitely has PSRAM
Upload Speed	921600 (reduce to 460800 or 115200 if unstable)
Port	Correct COM port of the ESP32-S3

4. Serial monitor settings

- Baud rate: 115200
- Line ending: No line ending
- Important: USB CDC On Boot should be Enabled on ESP32-S3 if serial output is expected over USB

5. Hardware / firmware expectations

- I2C pins expected by the firmware: SDA = GPIO 8, SCL = GPIO 9
- BLE advertised device name: MADEPLUS SMART SHOE
- BLE service UUID: 0000feed-0000-1000-8000-00805f9b34fb

6. Forwardable test note

Please test this firmware in Arduino IDE with the following setup:

Required installs:

1. esp32 by Espressif Systems
2. NimBLE-Arduino

Board settings:

- Board: ESP32S3 Dev Module
- USB CDC On Boot: Enabled
- CPU Frequency: 240 MHz
- Flash size: match board
- Partition Scheme: default / board default
- PSRAM: Disabled unless confirmed present
- Upload Speed: 921600 (reduce if unstable)
- Serial Monitor baud: 115200

No extra sensor libraries are needed for MPU6050, MS5611, or magnetometer because the code uses direct register-level I2C access.

Firmware expectations:

- BLE name: MADEPLUS SMART SHOE
- I2C pins: SDA = 8, SCL = 9